# Practice workbook · Deduce genotypes from a pedigree, family or blood group

Assigning genotypes to people in family trees, including X-linked and Y-linked symbols, possible ABO genotypes for a blood group, and matching children to parents.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEDUCE-GENOTYPES-FROM-PEDIGREE

# 1. Question 5 · part b.ii · 1 mark

November 2023 Paper 2

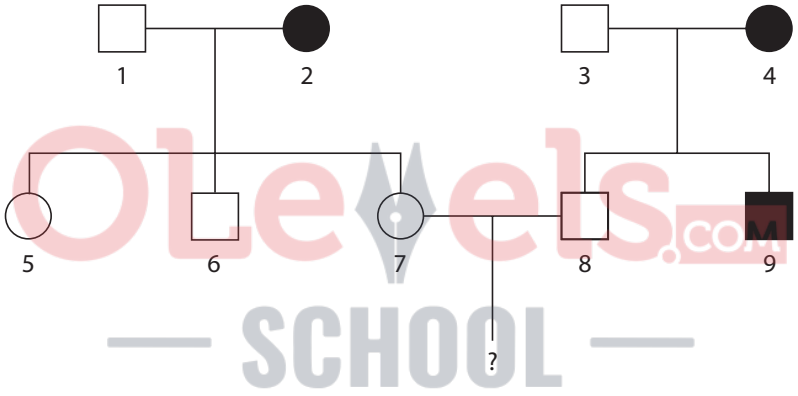
Worked-example retry · Attempt part b.ii only.

**5** Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

**(2)**

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.




**Key**

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

**(2)**

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

**(1)**

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P 7 4 4 9 4 A 0 1 2 2 0

# 1. Question 5 · part b.ii · 1 mark

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

**(4)**

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

**(4)**

probability = .....

**(Total for Question 5 = 13 marks)**

## 1. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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## 2. Question 9 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

**9** (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

**(3)**

(ii) Explain why people with PKU are advised to follow a low-protein diet.

**(3)**

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

**(3)**

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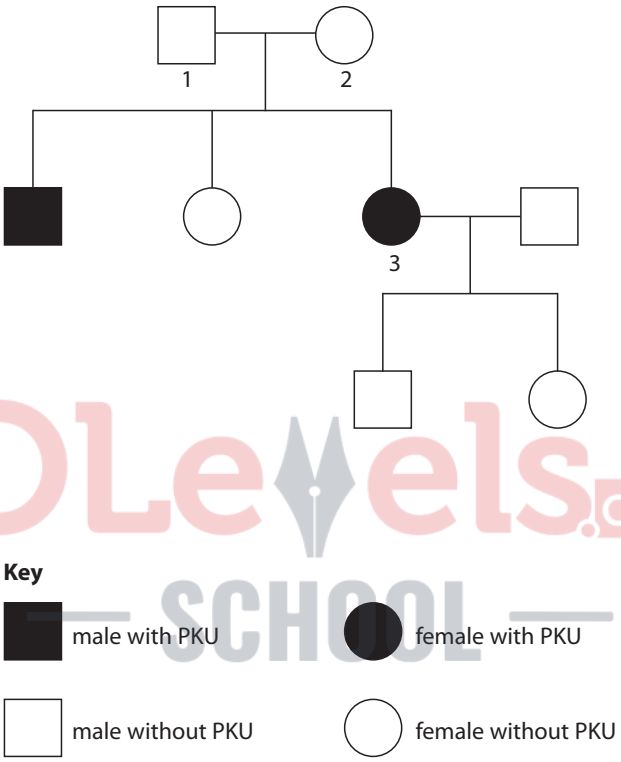
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## 2. Question 9 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

(b) The diagram shows how PKU is inherited in a family.




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(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

**(1)**

.....

(ii) Determine the probability that person 2 has one allele for PKU.

**(1)**

.....

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## 2. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

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## 3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

**9** (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

**(3)**

(ii) Explain why people with PKU are advised to follow a low-protein diet.

**(3)**

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

**(3)**

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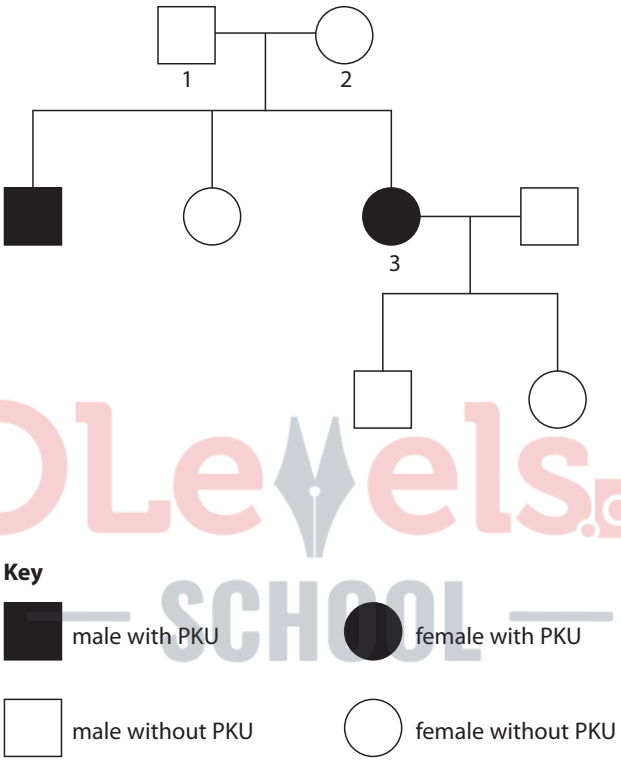
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# 3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

(b) The diagram shows how PKU is inherited in a family.




(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

**(1)**

(ii) Determine the probability that person 2 has one allele for PKU.

**(1)**

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## 3. Question 9 · part b.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Deduce the genotype of person 3.

You may draw a diagram to support your answer.

**(3)**

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**(Total for Question 9 = 14 marks)**

Working and answer

# 4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

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**9** (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

**(3)**

(ii) Explain why people with PKU are advised to follow a low-protein diet.

**(3)**

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(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

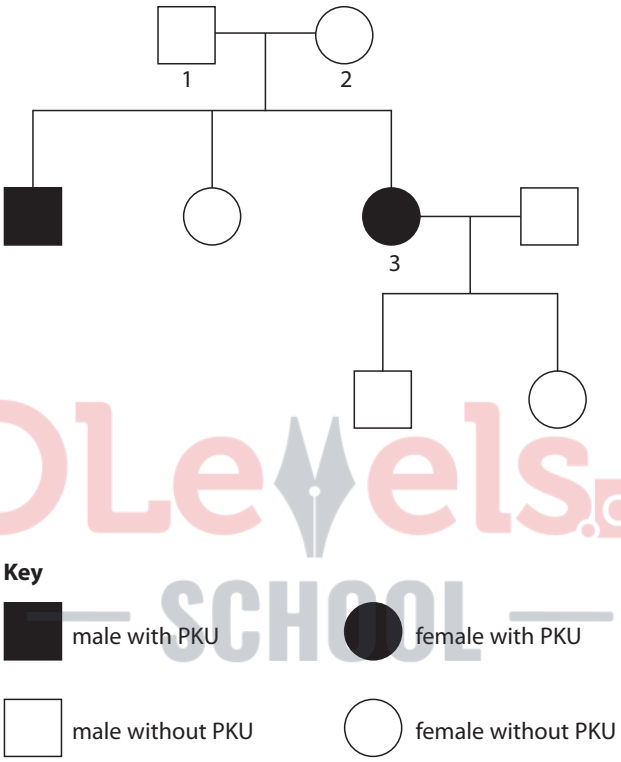
**(3)**

# 4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

(b) The diagram shows how PKU is inherited in a family.




(i) The allele for PKU can be represented by n. The normal allele can be represented by N.

Give the genotype of person 1.

**(1)**

(ii) Determine the probability that person 2 has one allele for PKU.

**(1)**

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# 4. Question 9 · part b.iii · 3 marks

January 2020 Paper 2

Further practice · Attempt part b.iii only.

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(iii) Deduce the genotype of person 3.

You may draw a diagram to support your answer.

**(3)**

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**(Total for Question 9 = 14 marks)**

## 4. Question 9 · part b.iii · 3 marks · Working and answer

Working and answer

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## 5. Question 6 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

**6** Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

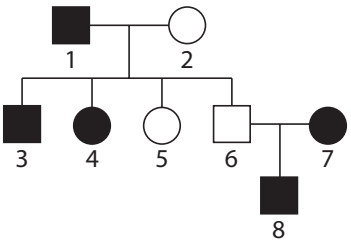
**(3)**

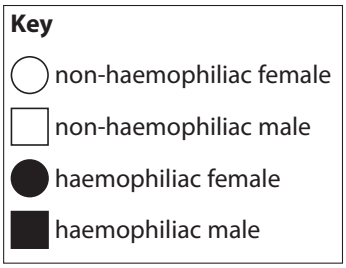
(ii) State what is meant by the term **sex-linked**.

**(2)**

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.




$X^h$ is the allele for haemophilia and $X^H$ is the allele for clotting.

(i) What is the genotype of person 3?

**(1)**

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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## 5. Question 6 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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(ii) What is the genotype of person 5?

**(1)**

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

**(2)**

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probability = .....

(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

**(3)**

.....

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**(Total for Question 6 = 12 marks)**

## 5. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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# 6. Question 6 · part b.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

**6** Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

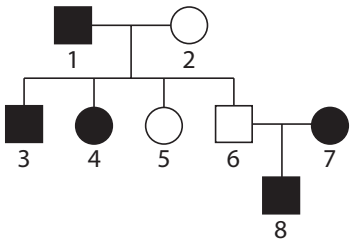
**(3)**

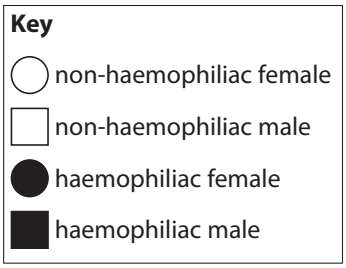
(ii) State what is meant by the term **sex-linked**.

**(2)**

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.




$X^h$ is the allele for haemophilia and $X^H$ is the allele for clotting.

(i) What is the genotype of person 3?

**(1)**

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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## 6. Question 6 · part b.ii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) What is the genotype of person 5?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = .....

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(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

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**(Total for Question 6 = 12 marks)**

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## 6. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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# 7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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**6** Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

**(2)**

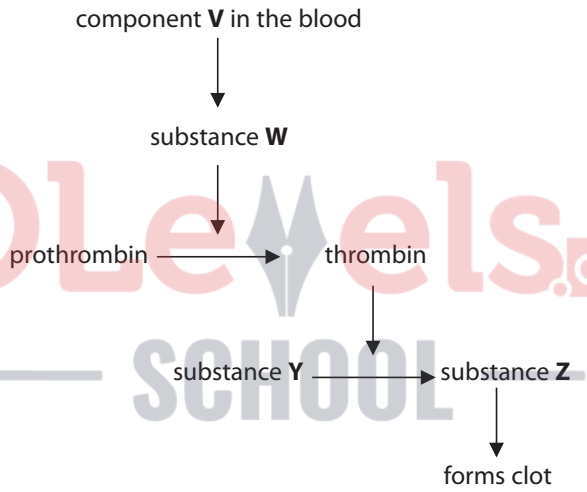
1 .....

.....

2 .....

.....

(ii) The diagram shows some of the stages in the process of blood clotting.




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Use words from the box to identify V, W, Y and Z.

| | | | | |
|---|---|---|---|---|
| thrombokinase enzyme | | white blood cells | | platelets |
| haemoglobin | fibrinogen | fibrin | antigens | antibody |

**(4)**

V .....

W .....

Y .....

Z .....



P 7 0 9 5 0 A 0 1 7 2 4

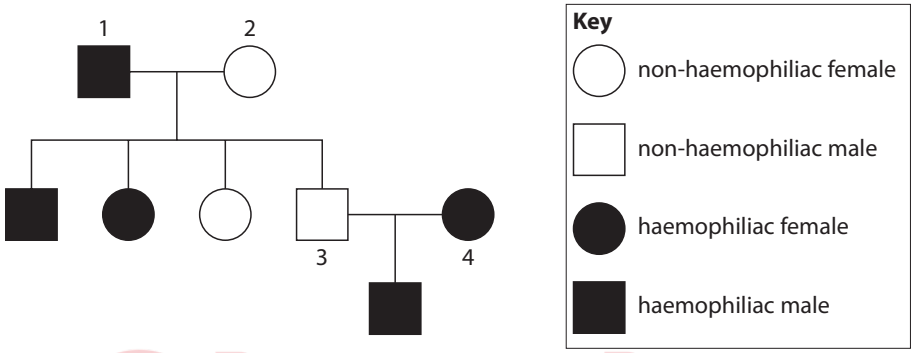
# 7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.




(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer. **(5)**

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## 7. Question 6 · part b.i · 5 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(ii) Male 3 and female 4 have a second child.

Deduce the probability that this child will be a boy with haemophilia showing how you obtained your answer.

**(4)**

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**(Total for Question 6 = 15 marks)**

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## 7. Question 6 · part b.i · 5 marks · Working and answer

Working and answer

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# 8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

**7** People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

**(3)**

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

**(2)**

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**20**

# 8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

**(2)**

percentage = ..... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

**(2)**

1 .....

.....

2 .....

.....

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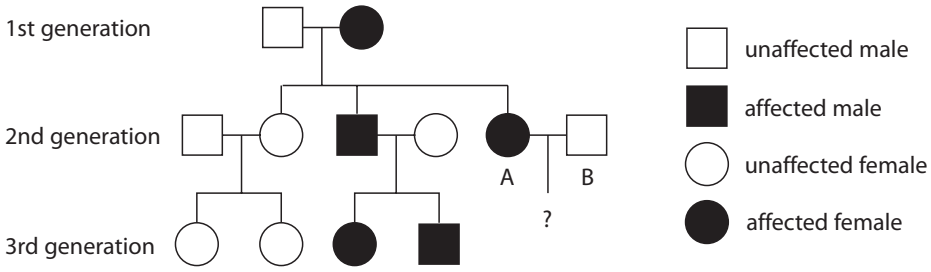
## 8. Question 7 · part c.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part c.i only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.




(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

**(1)**

ratio = .....

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## 8. Question 7 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

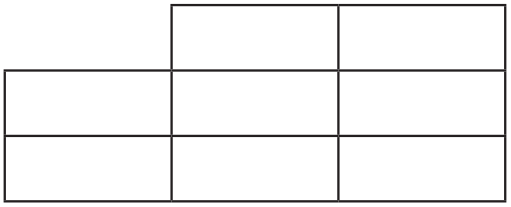
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(ii) Explain the possible genotypes and phenotypes for the offspring of individual A and individual B.

You should complete the genetic diagram as part of your answer.

**(4)**



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**(Total for Question 7 = 14 marks)**

Working and answer

## 9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

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**6** Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

**(4)**

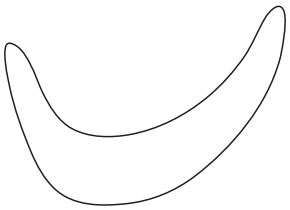
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## 9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

**(5)**

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## 9. Question 6 · part b · 4 marks

June 2023 Paper 2

Further practice · Attempt part b only.

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(b) A woman, heterozygous for sickle cell anaemia, and a man of unknown genotype for sickle cell anaemia have a child. The child is born with sickle cell anaemia.

Explain the possible genotypes of the man.

Use H to represent the allele for unaffected haemoglobin and h to represent the allele for sickle haemoglobin.

**(4)**

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**(Total for Question 6 = 13 marks)**

## 9. Question 6 · part b · 4 marks · Working and answer

Working and answer

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# 10. Question 5 · part b.i · 2 marks

November 2023 Paper 2

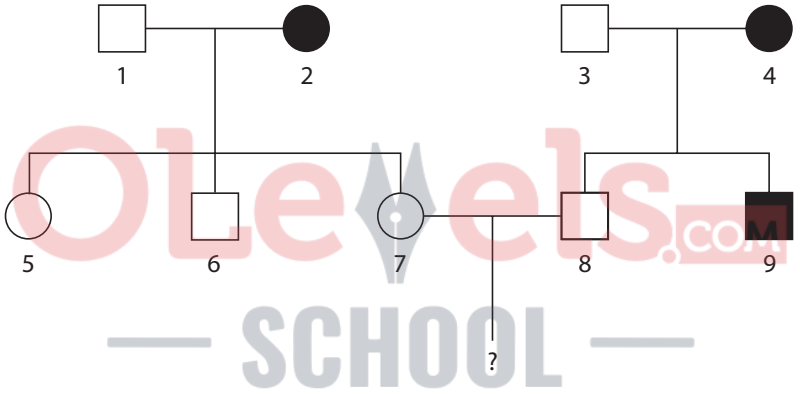
Further practice · Attempt part b.i only.

**5** Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

**(2)**

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.




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**Key**

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

**(2)**

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

**(1)**

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## 10. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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# 11. Question 4 · part c.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.i only.

**4** (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

| Blue | Green | Brown | Hazel | Grey |
|---|---|---|---|---|
| 𝍸 𝍸 𝍸 𝍸 𝍸 | 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 | III |

(i) Complete the table by giving the missing information.

**(1)**

| Number of people with each eye colour | | | | |
|---|---|---|---|---|
| **Blue** | **Green** | **Brown** | **Hazel** | **Grey** |
| 25 | | | | 3 |

(ii) Calculate the percentage of people with brown eyes in the sample.

**(2)**

percentage = ..... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

**(1)**

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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P 7 4 4 9 5 A 0 1 2 2 8

## 11. Question 4 · part c.i · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c.i only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

**(4)**

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

**(2)**

(ii) Give one symptom of cystic fibrosis.

**(1)**

**(Total for Question 4 = 11 marks)**

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## 11. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

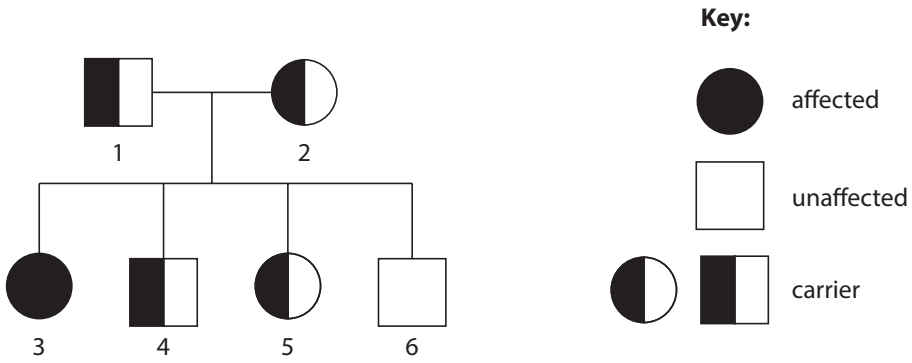
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## 12. Question 8 · part a.i · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.i only.

**8** (a) The diagram shows the inheritance of cystic fibrosis in a family.




(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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## 12. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

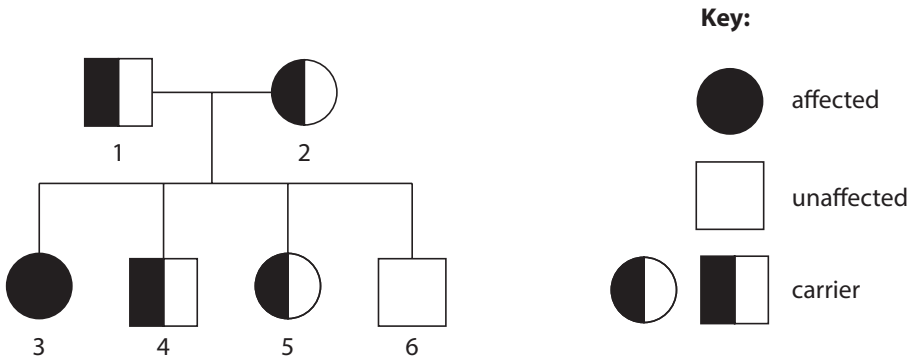
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## 13. Question 8 · part a.ii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.ii only.

**8** (a) The diagram shows the inheritance of cystic fibrosis in a family.




(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

**(1)**

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

**(1)**

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

**(2)**

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## 13. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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# 14. Question 7 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

**7** (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

| | Blood group | |
|---|---|---|
| | **mother** | **father** |
| 1st set of parents | AB | O |
| 2nd set of parents | B | B |

(i) Explain which child belongs to which set of parents.

**(4)**

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(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

**(3)**

probability = .....

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## 14. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

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## 15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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**5** (a) (i) Describe what is meant by the term **mutation**.

**(2)**

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

**(3)**

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## 15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

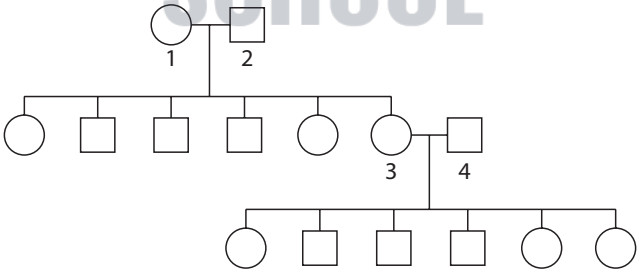
Further practice · Attempt part b.ii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

**(2)**

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.




Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

**(2)**

person 1

person 2

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DO NOT WRITE IN THIS AREA

## 15. Question 5 · part b.ii · 2 marks

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Neither person 3 nor person 4 has the condition.

Explain why the probability of their next child being a girl without hairy ears is 50%.

**(3)**

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**(Total for Question 5 = 12 marks)**

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Working and answer